

Thiago Vasco

Vancouver BC, Canada | Mobile: (604) 362-4131 | Email: thiagomscdv@gmail.com | Website: <https://thiagodcv.github.io/>

EDUCATION

University of British Columbia

Present

Master of Science, Mathematics, Institute of Applied Mathematics (IAM), Cumulative Average 88.6%

- Writing a thesis partially funded by the Burnaby oil refinery on how to make LP-MPC controllers more interpretable in the context of large industrial processes
- Co-supervised by Prof. Philip Loewen and Prof. Bhushan Gopaluni

University of British Columbia

Graduated: 12/23

Bachelor of Science, Major in Statistics, Minor in Mathematics, Cumulative Average 85.3%

- UBC Centennial Scholars Entrance Award, Dean's Honour List, British Columbia Government Scholarship, Professor Richard Lipsey Award in International Economics

EMPLOYMENT

Intern Analyst, Applied Research & Models, CPP Investments, Toronto ON

05/22 – 08/22

- Developed knowledge of professional software engineering practices for a full software development cycle, including continuous integration, coding standards, and code reviews
- Added functionality to analyze factor and cash exposures within asset allocation software
- Helped research a novel factor investing method which better characterizes idiosyncratic risk of illiquid asset classes
- Handled other ad hoc requests such as data cleaning and updating software packages

Intern Analyst, Portfolio Design & Construction, CPP Investments, Toronto ON

01/22 – 04/22

- Debugged and added functionality to experimental dynamic portfolio optimization model created by the team
- Modified and ran regressions for risk premia forecasting and parameter estimation
- Created a high-quality GUI using R Shiny for researching optimal factor exposures under different model assumptions

Project Assistant, CANARY, UBC Vancouver Stroke Program, Vancouver BC

05/21 – 08/21

- Developed software for training and testing of Hugging Face transformer models applied to NLP classification, and contrasted performance with simpler models and other types of neural architectures
- Wrote code to parse language data and to accomplish other preprocessing tasks

CLASS PROJECTS

LQR Missile Guidance (MECH 509 – Control Theory)

- Implemented an LQR-based control strategy for controlling a nonlinear, time-varying system using Python

Primal-Dual Interior-Point Methods with Application to MPC (CPSC 536M – Convex Analysis and Optimization)

- Implemented two primal-dual interior-point methods using Julia for solving linear and nonlinear problems in optimal control

Model-Based Reinforcement Learning Using Predictive Sampling MPC (CPSC 533V – Learning to Move)

- Implemented a model-based RL algorithm where system dynamics are modeled using a neural network, and optimal actions are found by using a sampling-based MPC approach

ADDITIONAL INFORMATION

- **Legal Status:** Canadian Citizen
- **Academic Interests:** Control, Optimization, Reinforcement Learning, Machine Learning
- **Languages & Frameworks:** Python, PyTorch, Java, C, C++, R, Julia, MATLAB, GitHub, Jira, Confluence, LATEX, and others
- **Relevant Courses:** Calculus I-IV, Linear Algebra, Probability, Statistics, Machine Learning, Time Series, Differential Equations, Control Theory, Numerical Optimization, OOP Software Engineering, Algorithms